Eric Zhang's Physics Notes

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Preface

This document was created as a collection and expansion of my physics notes from AP Physics C: Electricity and Magnetism, AP Physics C: Mechanics, the University of Maryland's PHYS273: Intermediate Oscillations and Waves. Note that the classes are listed in the order that I took them. However, the notes are arranged with electricity and magnetism *after* mechanics, since that is the order that most people take them in.

As a result of electricity and magnetism being my first physics class, the "Electricity and Magnetism" section has relative paucity of content compared to the "Mechanics" and "Waves and Oscillations" section.

These notes are *not* a replacement for an actual physics textbook and were *never* meant to do so.

These notes are structured in a way to separate physics and mathematical techniques. Each chapter contains a list of the recommended mathematical topics, including links to the relevant sections.

Chapter 1

Mechanics

Chapter 2

Electricity and Magnetism

Recommended mathematical topics: A.1

2.1 Maxwell's Equations

Chapter 3

Waves and Oscillations

Recommended mathematical topics: A.2

3.1

Chapter 4

Special and General Relativity

4.1 Special Relativity

4.2 General Relativity

Chapter 5

Quantum Mechanics

5.1 Schrödinger's Equation

Appendix A

Mathematics and Notation

A.1 Vectors

A.2 Complex and Imaginary Numbers

References